FIT5230 Malicious AI

Generative Diffusion Model

Before we begin…



Reference: “Théâtre D’opéra Spatial” entry for the Colorado State Fair.

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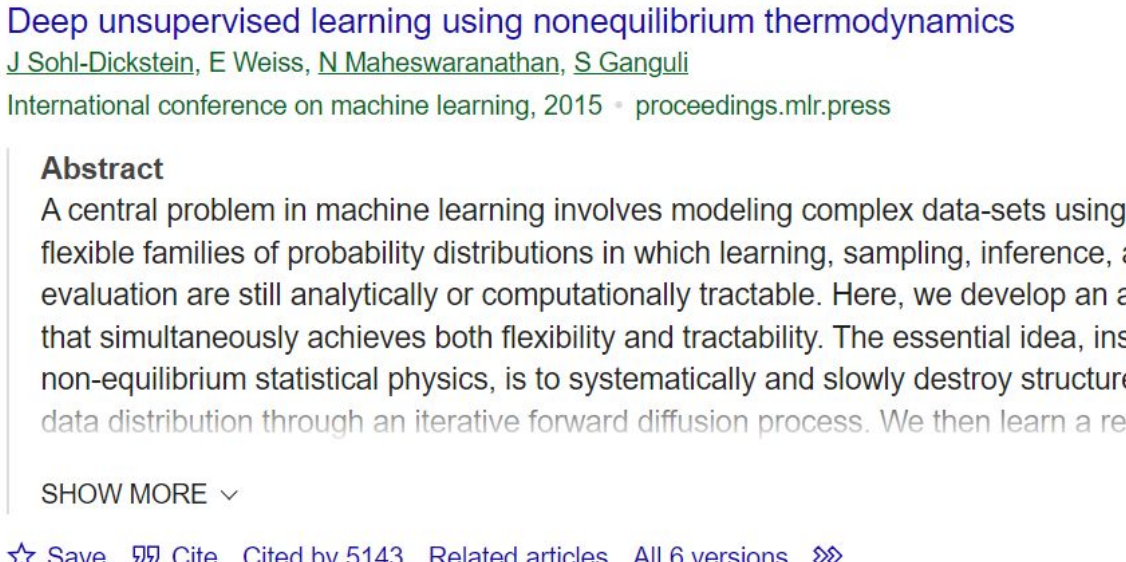
Generative Diffusion Model
Introduction

Generative Diffusion Model

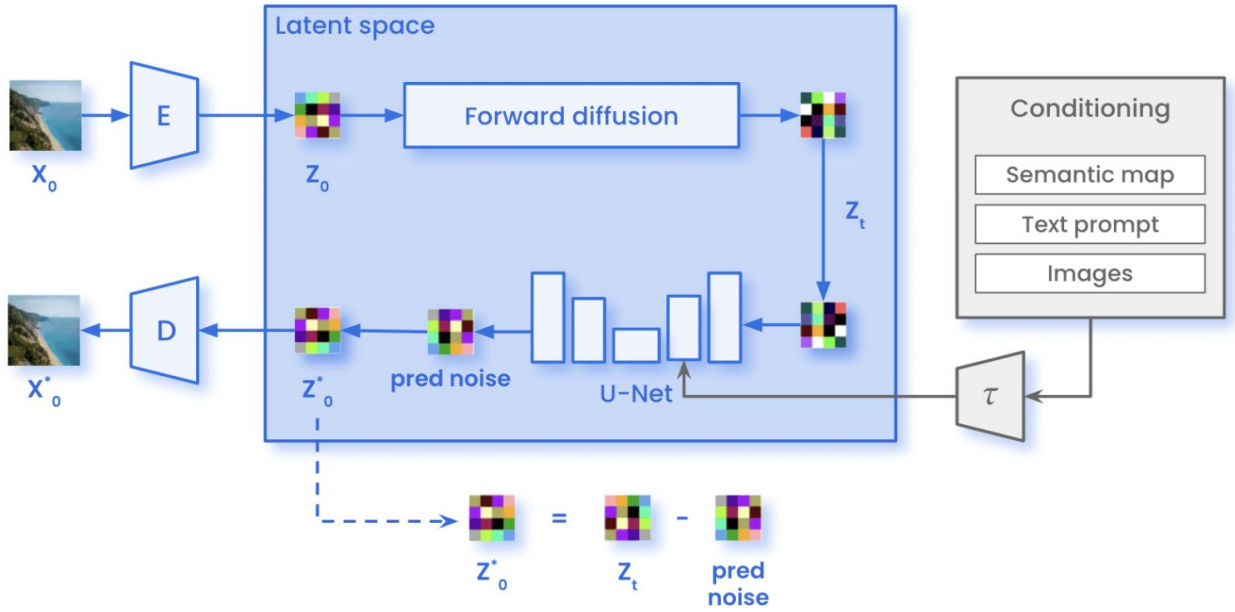
- Aim: To generate high-quality data by progressively adding noise to a dataset and then learning to reverse this process.
- Idea: gradually degrading data quality, only to reconstruct it to its original form or transform it into something new.

Two main processes:

- Forward diffusion process
- Reverse diffusion process



Generative Diffusion Model



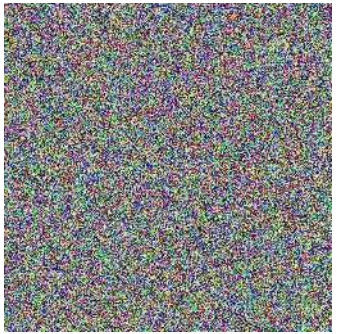
Reference: https://blog.marvik.ai/2023/11/28/an-introduction-to-diffusion-models-and-stable-diffusion/

GDM Forward Diffusion Process

- Introduce noise into the input



Original Image

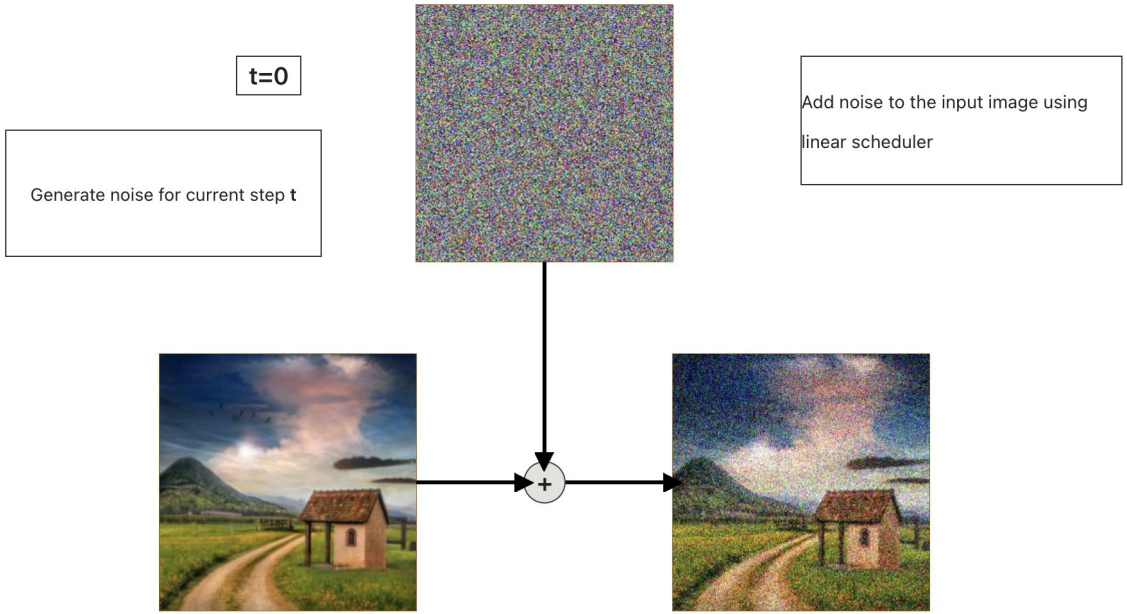


Pure noise

- Any image, e.g., the original image, has non-random distribution.
- Since the distribution is unknown, the distribution is destroyed by adding noise to it → output = pure noise

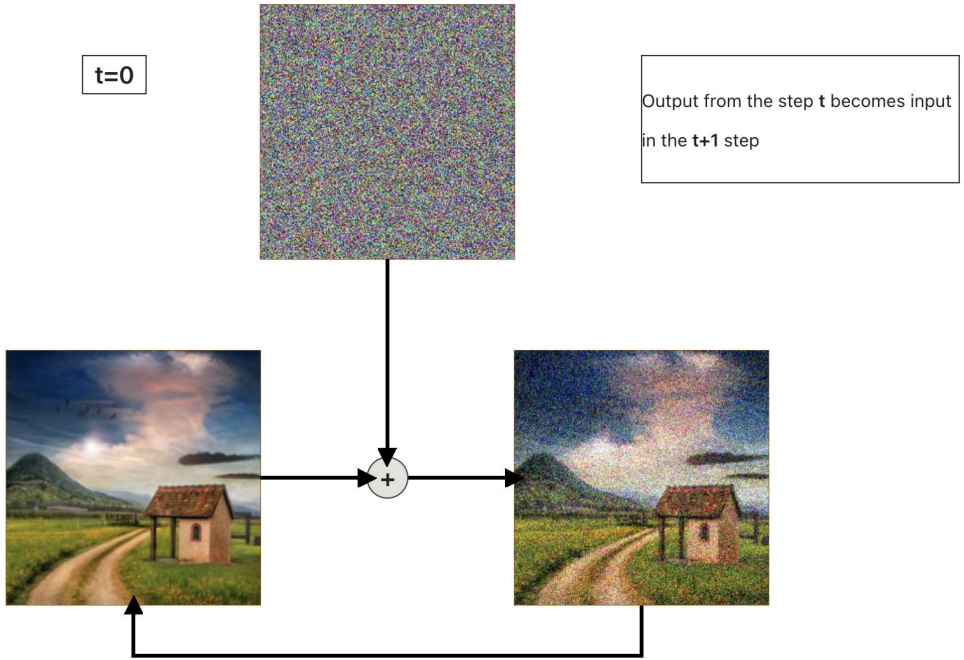
Reference: https://erdem.pl/2023/11/step-by-step-visual-introduction-to-diffusion-models

GDM Forward Diffusion Process



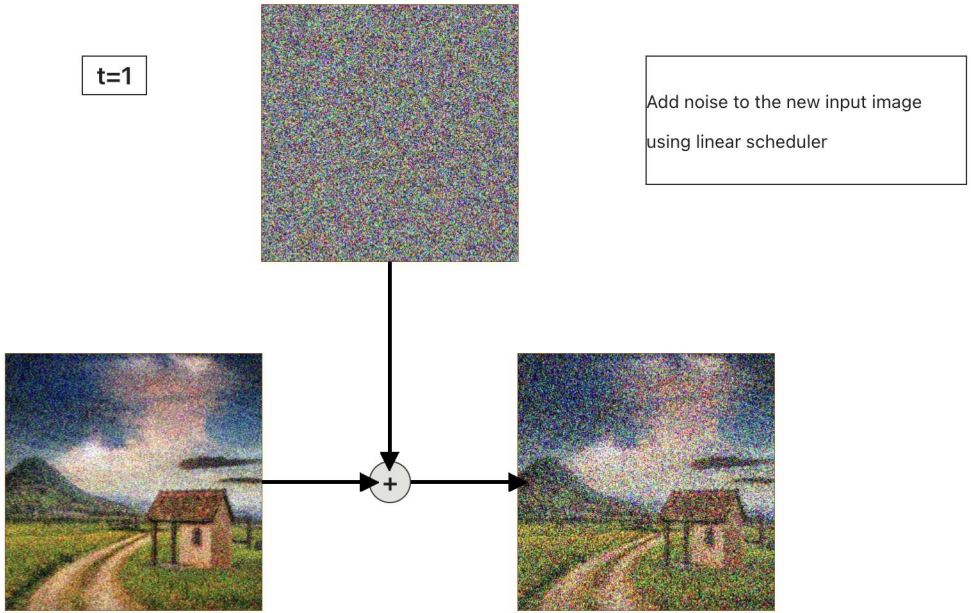
Reference: https://erdem.pl/2023/11/step-by-step-visual-introduction-to-diffusion-models

GDM Forward Diffusion Process



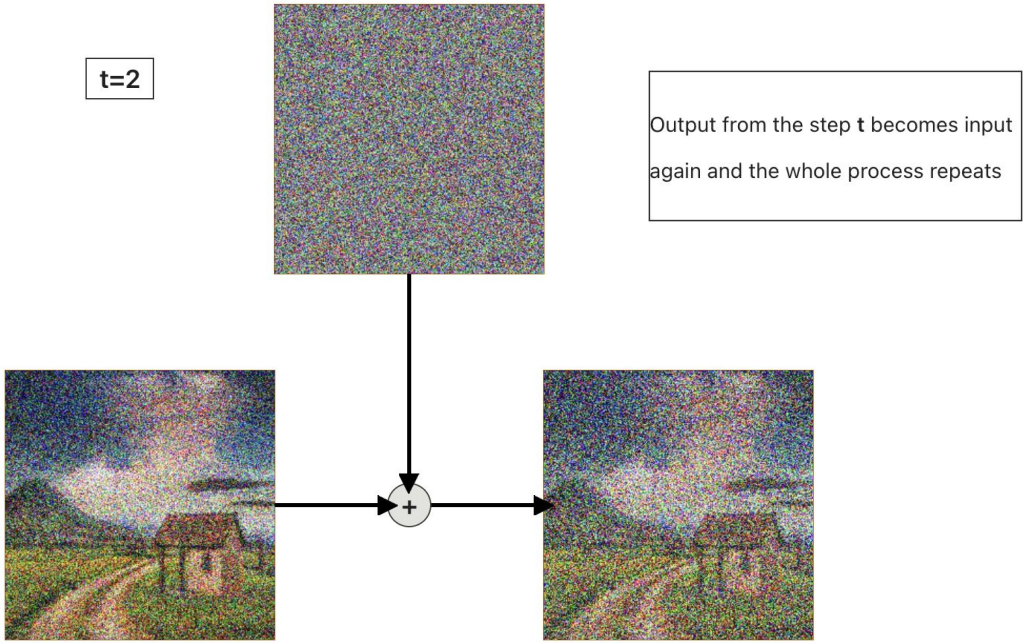
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GDM Forward Diffusion Process



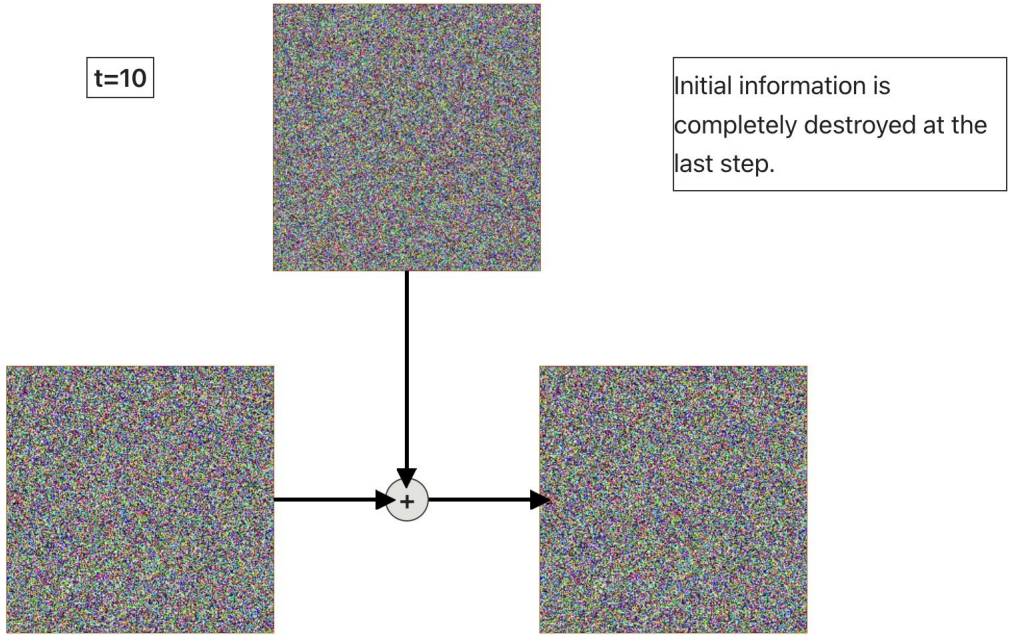
Reference: https://erdem.pl/2023/11/step-by-step-visual-introduction-to-diffusion-models

GDM Forward Diffusion Process



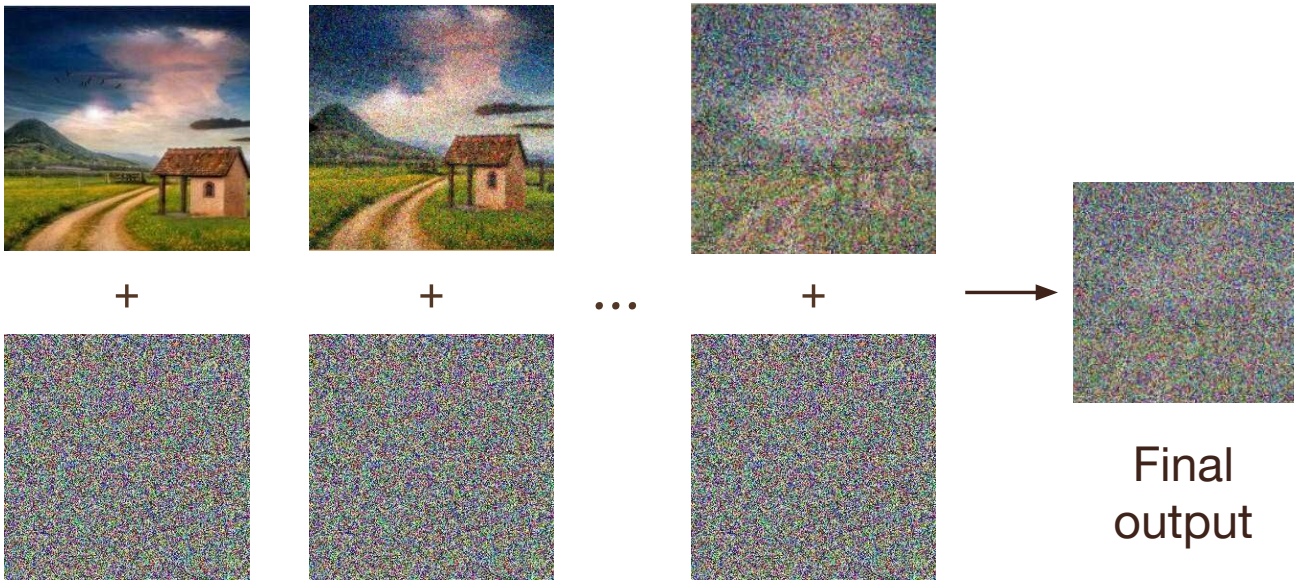
Reference: https://erdem.pl/2023/11/step-by-step-visual-introduction-to-diffusion-models

GDM Forward Diffusion Process



Reference: https://erdem.pl/2023/11/step-by-step-visual-introduction-to-diffusion-models

GDM Forward Diffusion Process



GDM - Noise Scheduler

- The noise scheduler, βt, act like percentages for the amount of noise added at time t relative to time t-1.
- αt_bar is a product of all αt values from 0 to t.
- The noise added at time t is the product of all αt values.
- The amount of noise at each timestep increases exponentially, and the percent of the original image decreases exponentially.

$$\alpha_t = 1 - \beta_t$$

$$\bar{\alpha}_t = \prod_{s=1}^{t} a_s$$

GDM - Noise Scheduler

A common choice for the noise schedule is to use a geometric progression:

$$\beta t = \beta * (1 - \beta)^{(T - 1 - t)}$$

where:

β is a constant between 0 and 1
T is the total number of steps

The number of steps T = the length of the forward and reverse diffusion processes

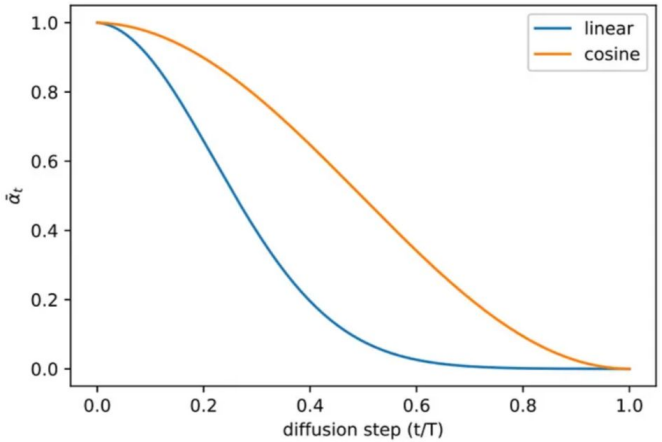
T and noise ↑
difficulty to recover from noise ↑
variation in the data ↑
T and noise ↓
difficulty to recover from noise ↓
variation in the data ↓

GDM - Linear Scheduler

- Linear schedule: 10^{-4} and 0.02
  - At time t=0, the value of βt will be 10^{-4}
  - At time T, β_t will be 0.02.

- Problem with the linear scheduler:
  - Small image is not far from pure Gaussian noise too early in the diffusing process
  - Noise is being added **too fast**
  - **Hard to learn** the reverse process.

GDM - Cosine Scheduler



Cosine schedule:

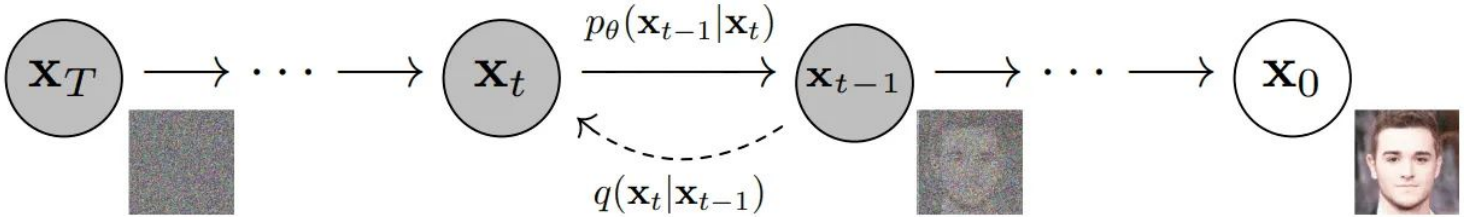
- add noise slower to retain image information for later timesteps



Reference: [Alex Nichol et al., Improved Denoising Diffusion Probabilistic Models, 2021]

GDM Reverse Diffusion Process

Diffusion model predicts the entire noise to be removed in a given timestep



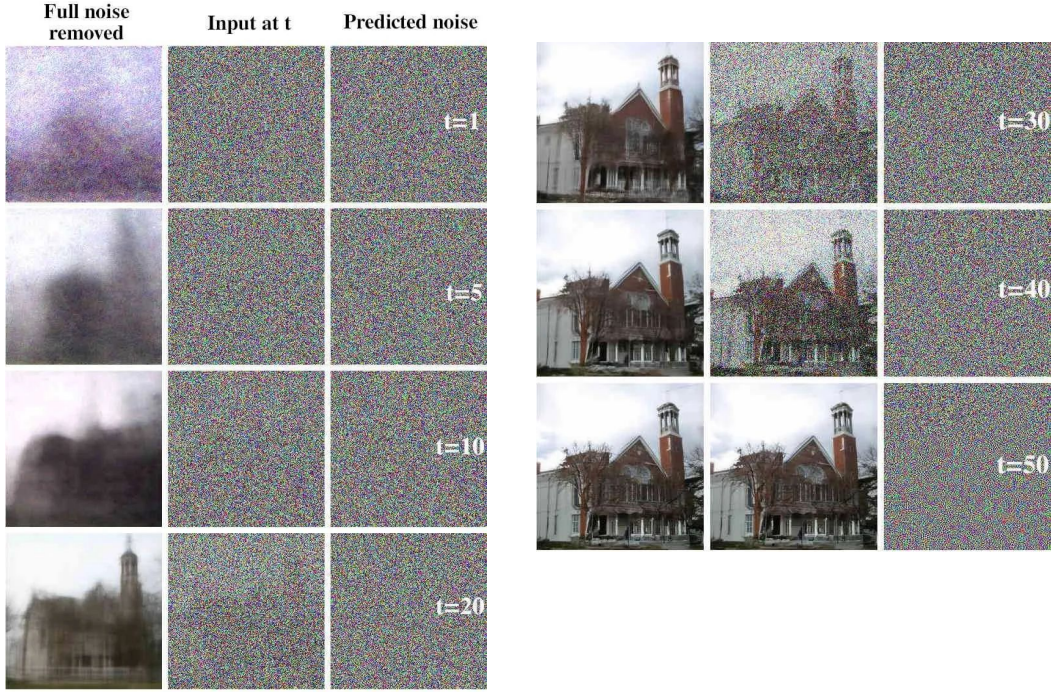
The output value for a given time step t can be computed:

$$x_{t-1} = \frac{1}{\sqrt{a_t}}(x_t - \frac{\beta_t}{\sqrt{1-\bar{\alpha}}}\epsilon_\theta(x_t, t)) + \sqrt{\beta_t}\epsilon$$

where $\epsilon_\theta(x_t, t)$ is our model's output (predicted noise)

Reference: [Deep Unsupervised Learning using Nonequilibrium Thermodynamics, arXiv:1503.03585, 2015.]

GDM Reverse Diffusion Process

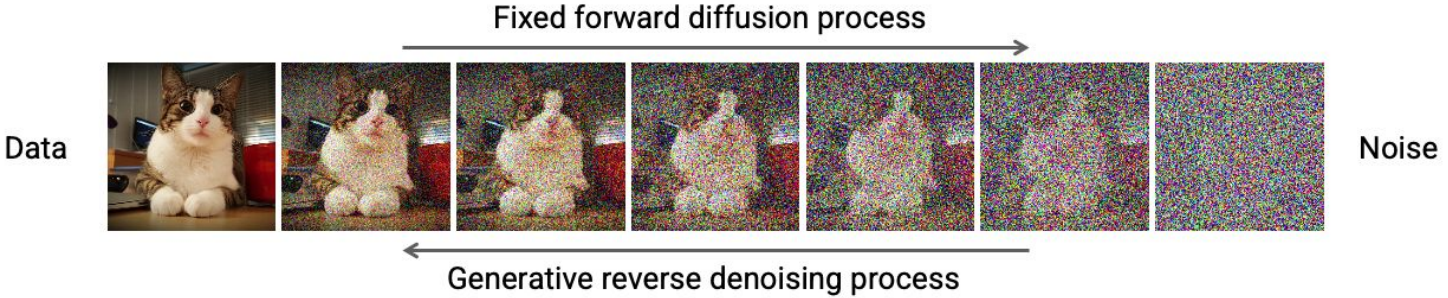


Reference: https://erdem.pl/2023/11/step-by-step-visual-introduction-to-diffusion-models

GDM Recap

Denoising Diffusion Models

- Learning to generate by denoising
- Denoising diffusion models consist of two processes:
  - Forward diffusion process that gradually adds noise to input
  - Reverse denoising process that learns to generate data by denoising

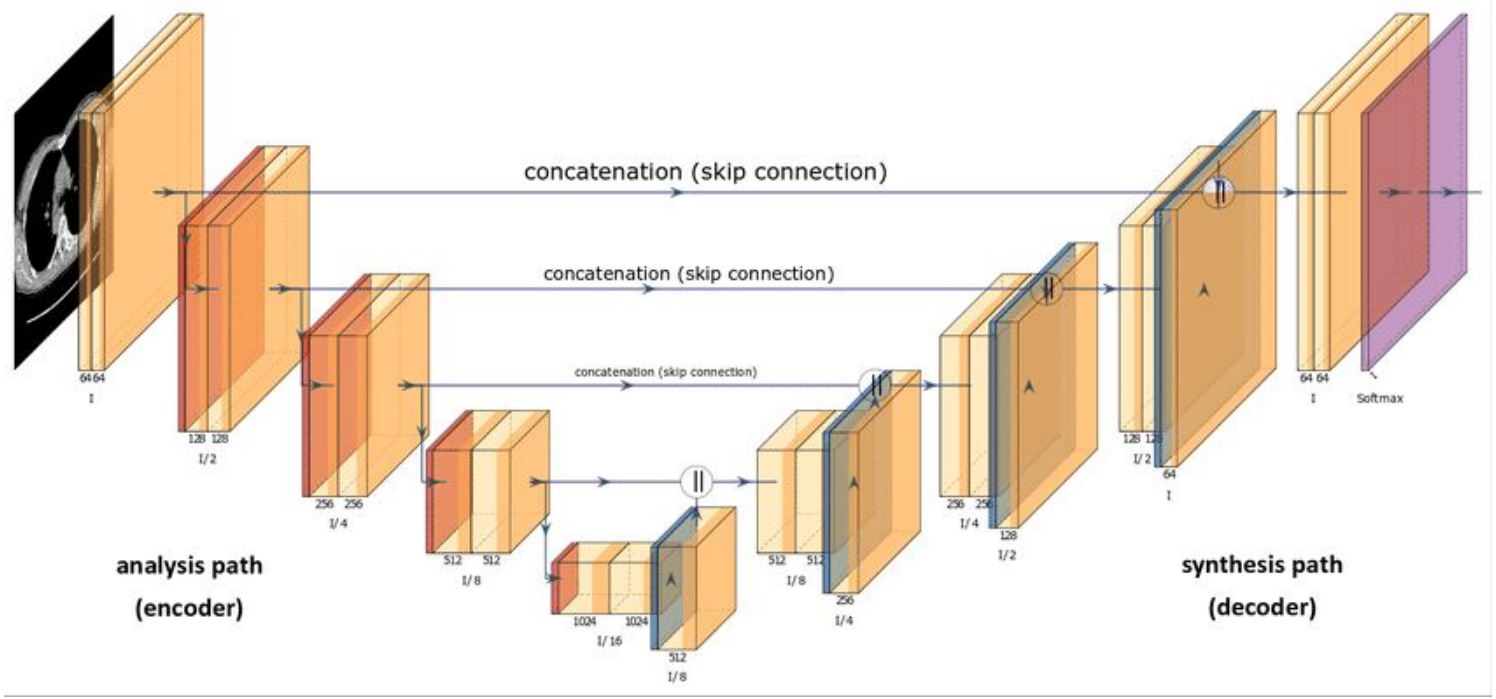


Reference: [Deep Unsupervised Learning using Nonequilibrium Thermodynamics, arXiv:1503.03585, 2015.]

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Generative Diffusion Model
Implementation

GDM Neural Network Architecture

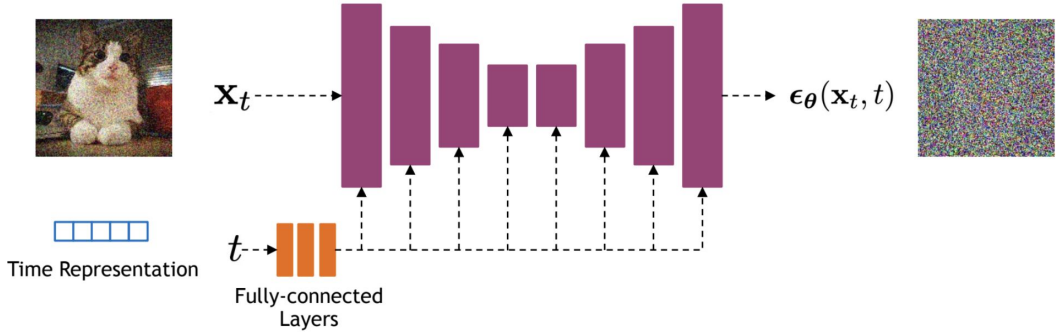


Reference: [Deep Unsupervised Learning using Nonequilibrium Thermodynamics, arXiv:1503.03585, 2015.]

GDM Implementation

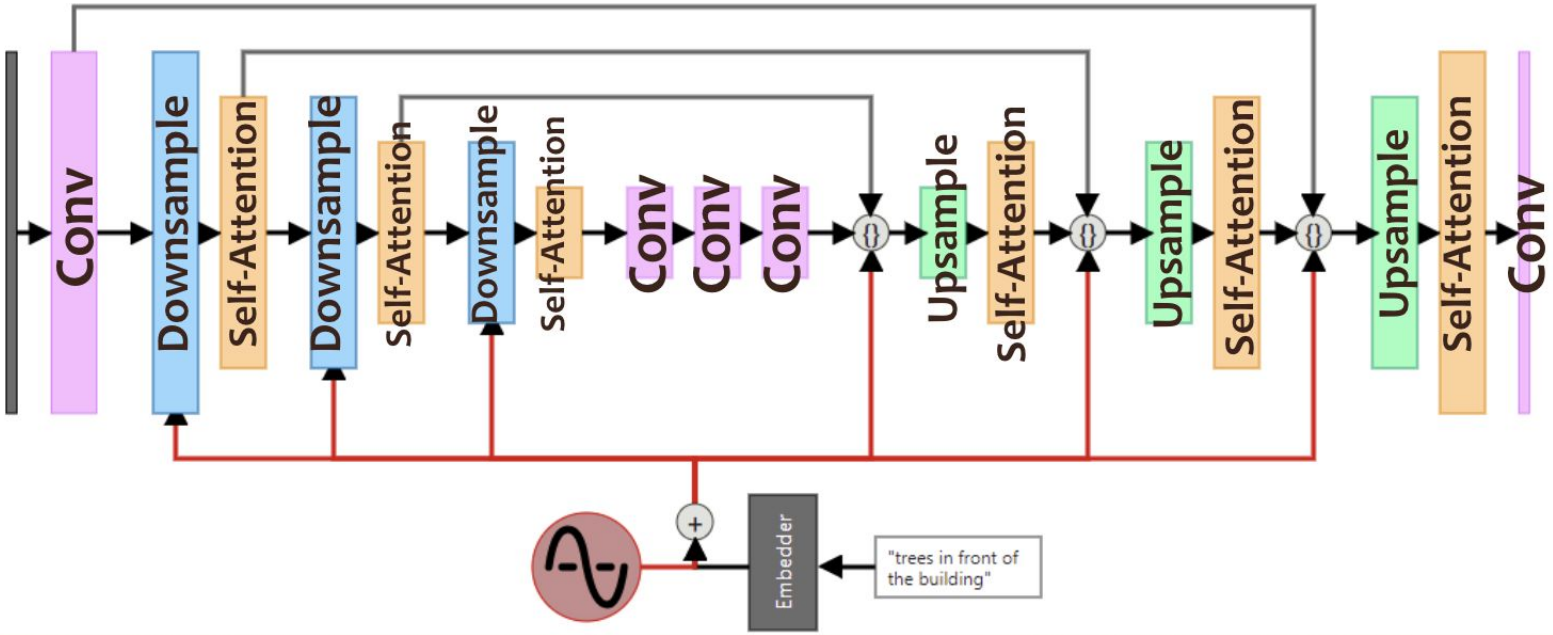
Diffusion models often use U-Net architectures with ResNet blocks and self-attention layers to represent.

- Time representation: **sinusoidal** positional embeddings or random Fourier features.
- Time features are fed to the residual blocks using either simple spatial addition or using adaptive group normalization layers.



Reference: [Deep Unsupervised Learning using Nonequilibrium Thermodynamics, arXiv:1503.03585, 2015.]

GDM Implementation



Reference: [Deep Unsupervised Learning using Nonequilibrium Thermodynamics, arXiv:1503.03585, 2015.]

GDM Implementation

- Embedding vectors:
  - adds embedding at each step
  - use sinusoidal encoding for encoding timestep t,
  - embedder for the prompt → if the model supports prompting
  - e.g., CLIP
- Outputs are then passed into downsample and upsample blocks.

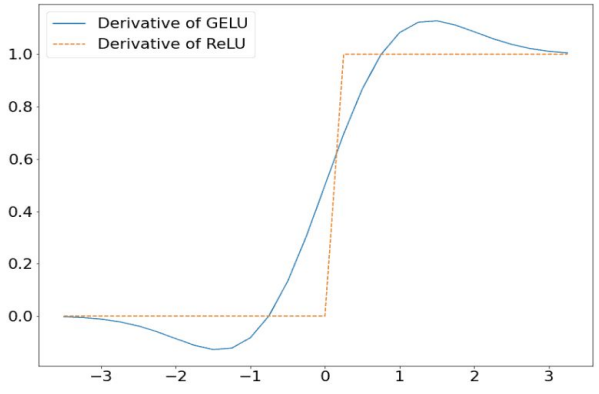
Reference: [Deep Unsupervised Learning using Nonequilibrium Thermodynamics, arXiv:1503.03585, 2015.]

GDM Modified U-Net

- Conv Block:
  - Simple ResNet block
  - Used in Upsample and Downsample blocks
- Gaussian Error Linear Unit (GELU):

$$\begin{aligned}\mathrm{GELU}(x) &= xP(X \leq x) = x\Phi(x) \\ &\approx 0.5x\left(1 + \tanh\left[\sqrt{2/\pi}\left(x + 0.044715x^3\right)\right]\right)\end{aligned}$$

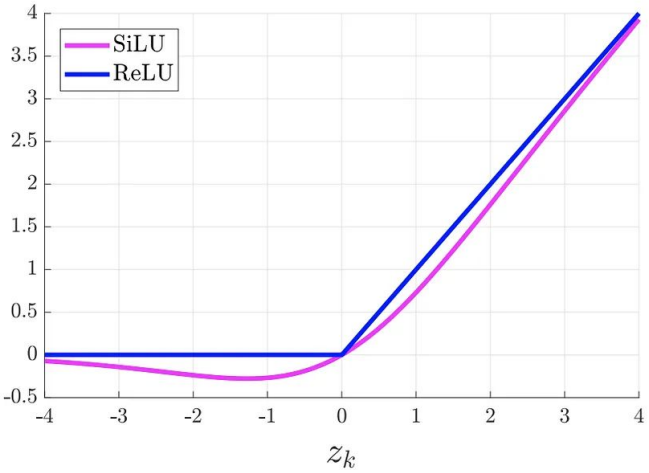
- ReLU activation function: hard cut-off at 0 for any negative number and otherwise produces a linear result.
- GELU follows a similar but smoother pattern.



Reference: [Deep Unsupervised Learning using Nonequilibrium Thermodynamics, arXiv:1503.03585, 2015.]

GDM Modified U-Net

- Downsample block:
  - Downsample input to the size of the next layer using MaxPool2d and 2 block of Conv.
- Embeddings are processed with a Sigmoid Linear Unit (SILU):
- SILU(x) = x * sigmoid(x)
- Both tensors are added and sent to the next block.



Reference: [Deep Unsupervised Learning using Nonequilibrium Thermodynamics, arXiv:1503.03585, 2015.]

GDM Modified U-Net

- Self-Attention Block
    - Multi-head self-attention is used

- Self-attention: Provides a context-aware representation for each element in a sequence.
- Multi-head self-attention: Learn **different aspects** of the relationships between elements
- E.g., relationships between text: grammar, word relationships

Reference: [Deep Unsupervised Learning using Nonequilibrium Thermodynamics, arXiv:1503.03585, 2015.]

GDM Modified U-Net

- Upsample block:
  - Takes Input from the previous layer and concatenate with residual connection (from the first ResNet block).
  - Embeddings are processed with a Sigmoid Linear Unit (SILU)
- All tensors are added and sent to the next block.

Reference: [Deep Unsupervised Learning using Nonequilibrium Thermodynamics, arXiv:1503.03585, 2015.]

GDM Forward Diffusion Process

- The loss function typically defined in diffusion models:

$$L_{DM} = \mathbb{E}_{\mathbf{x},t,\epsilon}\left[\|\boldsymbol{\epsilon} - \boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t)\|^2\right]$$

where

- ϵ = actual noise
- $\epsilon_\theta(x_t,t)$ = predicted noise

Reference: https://erdem.pl/2023/11/step-by-step-visual-introduction-to-diffusion-models

Denoising Diffusion Implicit Model (DDIM)

DENOISING DIFFUSION IMPLICIT MODELS

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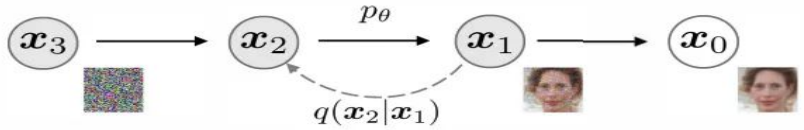
ABSTRACT

Denoising diffusion probabilistic models (DDPMs) have achieved high quality image generation without adversarial training, yet they require simulating a Markov chain for many steps in order to produce a sample. To accelerate sampling, we present denoising diffusion implicit models (DDIMs), a more efficient class of iterative implicit probabilistic models with the same training procedure as DDPMs. In DDPMs, the generative process is defined as the reverse of a particular Markovian diffusion process. We generalize DDPMs via a class of non-Markovian diffusion processes that lead to the same training objective. These non-Markovian processes can correspond to generative processes that are deterministic, giving rise to implicit models that produce high quality samples much faster. We empirically demonstrate that DDIMs can produce high quality samples $10\times$ to $50\times$ faster in

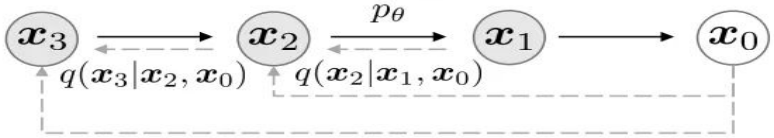
Reference: Denoising Diffusion Implicit Model (DDIM), ICLR 2021

Denoising Diffusion Implicit Model (DDIM)

- Speed of traditional DDPM process is too slow
  - takes 1,000 model passes to generate a single image
- The DDIM speed up image generation with little image quality tradeoff.
- The diffusion process is redefined as follows:



DDPM



DDIM

Reference: Denoising Diffusion Implicit Model (DDIM), ICLR 2021

Denoising Diffusion Implicit Model (DDIM)

- The DDIM allows skipping steps in the denoising process
  - Does not require all past states to be visited before the current state.
- The reverse diffusion process for a single step is redefined as follows:

$$x_{t-1} = \sqrt{\bar{\alpha}_{t-1}} \left(\frac{x_t - \sqrt{1-\bar{\alpha}_t}\epsilon_\theta^{(t)}(x_t)}{\sqrt{\bar{\alpha}_t}} \right) + \sqrt{1-\bar{\alpha}_{t-1}-\sigma_t^2} \cdot \epsilon_\theta^{(t)} + \sigma_t \epsilon_t$$

Reference: Denoising Diffusion Implicit Model (DDIM), ICLR 2021

GDM vs GAN

Image quality	↑	• Ease of training. • Simple and efficient loss functions. • Produces consistent, high-fidelity, often smoother images.	↓	• Can produce sharp, realistic images; prone to artifacts.
Training stability	↑	• Gradual data smoothing process. • Easier and more stable; likelihood-based optimization.	↓	• Mode collapse. • Hard to train; adversarial dynamics unstable.
Computational efficiency	↓	• Slow inference (many steps); improving with DDIM, Latent Diffusion. • Scales well with larger datasets/models.	↑	• Fast inference (single pass). • Hard to scale without instability.
Complexity	↑	• Complex theoretical underpinnings and implementation	↓	• Simple in theory, tricky in practice.

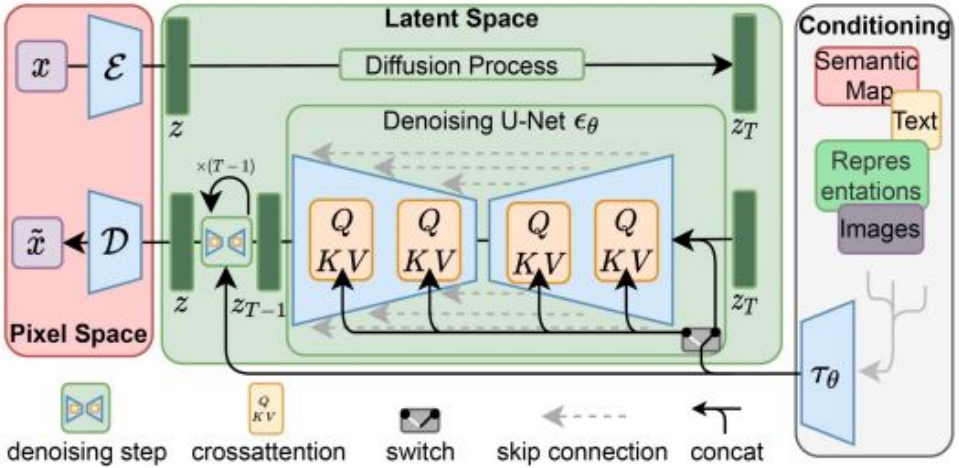
Reference: [Deep Unsupervised Learning using Nonequilibrium Thermodynamics, arXiv:1503.03585, 2015.]

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Deep Generative Model
Variational Autoencoder (VAE)

Latent Diffusion Model

1. Variational Autoencoder (VAE) based encoder and decoder
2. Denoising U-Net with self attention
3. Conditioning via cross attention



Reference: High-Resolution Image Synthesis With Latent Diffusion Models, CVPR 2022

Stable Diffusion Versions

1. SD 1.5: very common model, lightweight (860M parameters), lots of community support in the form of LoRAs, etc.
2. SD XL: same idea, just larger U-Net, cross-attention context, etc.
3. SD 3.0: no U-Net but Transformer-based backbone



Training data: subset of LAION-5B text-image dataset

Reference: SDXL: Improving Latent Diffusion Models for High-Resolution Image Synthesis, 2023
Code: https://github.com/Stability-AI/generative-models

Stable Video Diffusion & AnimateDiff

1. SVD: image-to-video generation model by Stability AI
2. AnimateDiff: adapter network for text-to-video generation

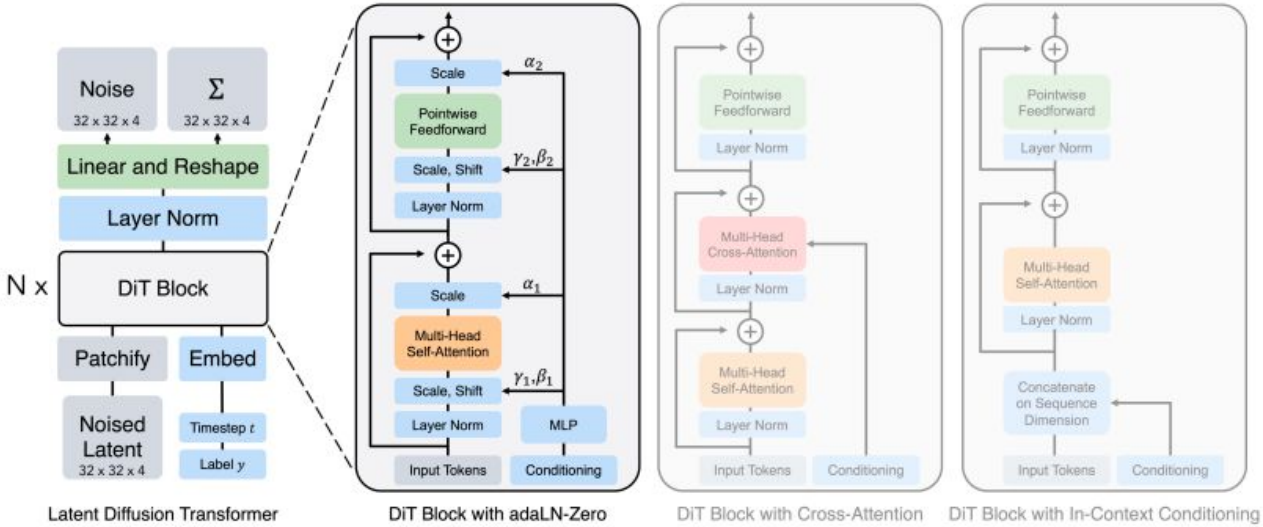
Both (and other video models) use (learned) temporal attention; only a few-second clips, about 30 frames total



Reference: Stable Video Diffusion: Scaling Latent Video Diffusion Models to Large Datasets, arxiv 2023
AnimateDiff: Animate Your Personalized Text-to-Image Diffusion Models without Specific Tuning", ICLR 2024

Diffusion Transformers

- Replace U-Net with vision transformer (need to add conditioning)
- Turns out to be more scalable than U-Net!



Reference: Scalable Diffusion Models with Transformers, ICCV 2023

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Generative Diffusion Model
Application

AudioPure: Defending against Adversarial Audio via Diffusion Model

ICLR 2023

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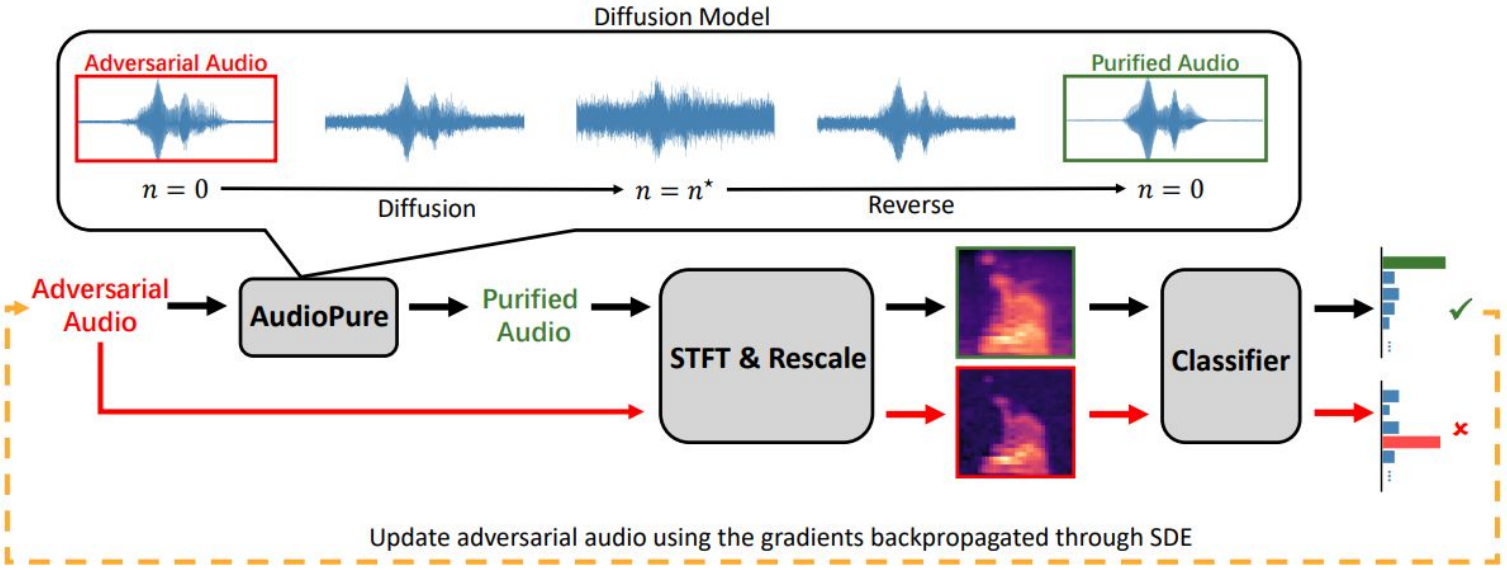
DEFENDING AGAINST ADVERSARIAL AUDIO VIA DIFFUSION MODEL

Shutong Wu[1,2*] **Jiongxiao Wang**[1] **Wei Ping**[3] **Weili Nie**[3] **Chaowei Xiao**[1]
[1]Arizona State University [2]Shanghai Jiao Tong University [3]NVIDIA

ABSTRACT

Deep learning models have been widely used in commercial acoustic systems in recent years. However, adversarial audio examples can cause abnormal behaviors for those acoustic systems, while being hard for humans to perceive. Various methods, such as transformation-based defenses and adversarial training, have been proposed to protect acoustic systems from adversarial attacks, but they are less effective against adaptive attacks. Furthermore, directly applying the methods from the image domain can lead to suboptimal results because of the unique properties of audio data. In this paper, we propose an adversarial purification-based defense pipeline, AudioPure, for acoustic systems via off-the-shelf diffusion models. Taking advantage of the strong generation ability of diffusion models, AudioPure first adds a small amount of noise to the adversarial audio and then runs the reverse sampling step to purify the noisy audio and recover clean audio. AudioPure is a plug-and-play method that can be directly applied to any pretrained classifier without any fine-tuning or re-training. We conduct extensive experiments on speech command recognition task to evaluate the robustness of AudioPure. Our method is effective against diverse adversarial attacks (e.g. $\mathcal{L}_2$ or $\mathcal{L}_\infty$-norm). It outperforms the existing methods under both strong adaptive white-box and black-box attacks bounded by $\mathcal{L}_2$ or $\mathcal{L}_\infty$-norm (up to +20% in robust accuracy). Besides, we also evaluate the certified robustness for perturbations bounded by $\mathcal{L}_2$-norm via randomized smoothing. Our pipeline achieves a higher certified accuracy than baselines. Code is available at `https://github.com/cychomatica/AudioPure`.

Defending against Adversarial Audio via Diffusion Model



Reference: Defending against Adversarial Audio via Diffusion Model, ICLR 2023
Code: https://github.com/cychomatica/AudioPure

Application of Popular Diffusion Tools

Midjourney

- Users can generate artwork and art concepts with Midjourney by using image generation commands or “prompts” through interactions with a bot on Discord chat.

DALL-E by Open AI

- DALL-E / DALL-E 2 can create realistic images and art from a description in a natural language, i.e., “prompts”.

Application of Popular Diffusion Tools

DreamStudio by Stability.ai

- based on the Stable Diffusion model,
- generating photo-realistic images given any text input.

Example Output

- Style: Cinematic
- Prompt: A young lady with silky red long hair reading a book under a tree



Application of Popular Diffusion Tools

Scenario:

- Allows users to create AI-generated game assets that are able to match a consistent look.
- Build and train your own AI engine in order to produce a style-consistent asset for your games.

Layer.ai:

- Allows users to create art and video game assets, especially for variations of an asset that matches a particular style.

Application of Popular Diffusion Tools

The first full AI animation movie: Uses Stable Diffusion



Reference: ‘Where the Robots Grow’ was the first AI animated feature film

Application of Popular Diffusion Tools

The first full AI music video with OpenAI's Sora



Reference: Director Paul Trillo Debuts World's First Sora-Powered Music Video, Unlocking Creative Possibilities 2024

Real-World Application Analysis

Your Mission:
You're consultants advising a company on implementing diffusion models in their business.

Choose Your Industry:
🎮 Gaming Studio (asset creation)
🎬 Animation Studio (character design)
📱 Marketing Agency (ad content)
🏗️ Architecture Firm (concept visualization)

Share your answer here: https://shorturl.at/pVlHz

Real-World Application Analysis

Task Breakdown:

1. Opportunity Analysis
   - What specific problems could diffusion models solve?
   - How does this compare to current methods?
2. Risk Assessment
   - What are the potential downsides?
   - Consider: quality, cost, ethics, legal issues
3. Recommendation
   - Should they adopt this technology? Why/why not?
   - What implementation strategy would you suggest?

Share your answer here: https://shorturl.at/pVlHz

Food for Thought 🤔

Reflection Questions:

1. How might this technology change job roles in your chosen industry?
2. What safeguards should be in place?

Share your answer here: https://shorturl.at/pVlHz